Introspective Imitation Dynamics and the Limits of Extrinsic Update Rules in Heterogeneous Public Goods Games

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Abstract

Evolutionary game theory provides a framework by which to study the emergence of cooperation in a population of self-interested actors. In such a framework, players' decision on whether or not to cooperate evolve according to decision rules called population dynamics. However, often games are studied under the assumption that all individuals play under the same conditions. In this paper, we explore the effect of different population dynamics in a heterogeneous public goods game. We consider extrinsic population dynamics which update players' actions according to the fitness of other players in the population. We also consider intrinsic population dynamics, in which players update their actions by considering their own fitness. We then introduce "introspective imitation dynamics", a novel population dynamic which combines both fitness-proportional extrinsic selection and self-referential intrinsic update steps. Finally, we shall see the limit on cooperation which is imposed by purely extrinsic population dynamics, and how it can be exceeded by considering intrinsic population dynamics.

1 Introduction

Reducing global carbon emissions is one of the defining collective action problems of our time. Countries differ substantially in their economic capacity, historical emissions, and the costs they face when decarbonising, yet they share a mutual interest in the global public good of a stable climate. This structure maps onto a *public goods game* [36, 13, 2, 1, 21]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor r and distributed equally. The temptation to free-ride means that purely selfish reasoning leads to a tragedy of the commons [11], even when collective cooperation would benefit everyone. Both theory [34] and controlled experiments [25, 24] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates. The challenge of overcoming free-riding is not merely theoretical. Nordhaus [27] proposed voluntary *climate clubs* in which member nations adopt a common carbon price and levy tariffs on non-members, turning the international climate problem into a coordination game with enforcement. The European Union's Carbon Border Adjustment Mechanism [7], which requires importers to purchase certificates reflecting the carbon content of goods produced under weaker environmental regulation, represents a real-world instance of this idea.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [14, 28]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner's dilemma [17, 3] and the snowdrift game [43]. The public goods game generalises these pairwise dilemmas to N players [12], making it well suited to studying multi-lateral cooperation such as climate agreements. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [30, 42].

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [13]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [29, 39, 31], heterogeneity generally promotes cooperation. In [35] it is shown that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of cooperation [33]. When contributions depend on dynamically accumulated reputation [23], high-reputation defectors receive reduced transfers, weakening defection's advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [19], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided r is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [8] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

The studies above are almost exclusively concerned with spatially structured populations, where network topology is central. Much less is known about fully heterogeneous, *well-mixed* populations, in which all individuals are distinguished by a fixed attribute (such as their contribution capacity) independently of any graph structure. Furthermore, most existing analyses rely on simulation or mean-field approximations; exact treatments of heterogeneous evolutionary dynamics remain rare. How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [26, 16], Fermi imitation dynamics [40], aspiration dynamics [6], and introspection dynamics [4, 5] all model strategy revision in distinct ways, and each may respond differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their own counterfactual payoff, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well. The introspection dynamics framework developed in [4] and extended to multiplayer asymmetric games in [5] formalises this distinction for homogeneous populations; extending it to the fully heterogeneous setting motivates the present work.

We make the following contributions. First, we establish a general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of N ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, we introduce a novel update rule, *Introspective Imitation Dynamics*, in which players select a role model proportionally to fitness (as in the Moran process) but evaluate whether to adopt that strategy by comparing their own current and counterfactual payoffs (as in introspection dynamics). This rule is well suited to heterogeneous populations, where copying a high-fitness individual is likely although irrational if the same strategy yields a worse payoff for the copying player. Third, we show that purely extrinsic dynamics are subject to a ceiling on the probability of cooperation that intrinsic dynamics can exceed, and we characterise the conditions under which this advantage arises, with the ratio of the public goods multiplier r to population size N as the central governing parameter. We study a *linearly heterogeneous* population, in which players' capacities are graded uniformly across the group.

The remainder of the paper is organised as follows. Section 2 introduces the population model, defines all five dynamics and their mutation variants, and specifies the heterogeneous public goods game. Section 3 presents our results, and supplementary material containing extended parameter sweeps is provided in Section 4.

2 The General Heterogeneous Population Dynamic Model

Following [4, 5], we define a general framework for evolutionary dynamics on a fully heterogeneous, well-mixed population.

2.1 Population and State Space

The population consists of N **ordered** individuals. Each individual selects an action from a common action set $A = (a_1, a_2, \ldots, a_k)$; we assume throughout that all individuals share the same action set of size k although this is not strictly required for introspection and aspiration dynamics [4, 6]. A *state* of the population is an ordered N-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_N) \in A^N$, and the *state space* S is the set of all such tuples. A payoff function $\pi\colon S \to \mathbb{R}^N$ assigns to each state a vector of individual payoffs; we write $\pi_i(\mathbf{a})$ for the payoff of individual i in state $\mathbf{a}$.

The population evolves as a Markov chain [37] on S. Transitions can only occur between states that differ in exactly one coordinate, that is, states $\mathbf{a}$ and $\mathbf{b}$ with Hamming distance $h(\mathbf{a}, \mathbf{b}) = 1$ [18]. We call the unique coordinate at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{a}, \mathbf{b})$ [5]. The set of states reachable from $\mathbf{a}$ in one step is $\mathrm{Neb}(\mathbf{a}) = \{\mathbf{b} \in S : h(\mathbf{a}, \mathbf{b}) = 1\}$ [5].

2.1.1 Graphical representation

If we consider a game with two actions, for example cooperate and defect, we can compare the graph of such a Markov chain to an N-dimensional hypercube. This is the graph whose node set consists of the 2^N N-dimensional boolean vectors, and which has edges between edges if and only if they differ in exactly one position [10]. The state space of an evolutionary game with two actions can be modelled in this way by denoting the strategies "0" and "1".

The graphical representation of a birth-death process differs from a hypercube in exactly two ways. First, a population may remain the same at a given step, and thus each vertex contains a self-loop. Additionally, the probability of transitioning from a state $\mathbf{a}$ to a state $\mathbf{b}$ is not necessarily the same as the probability of transitioning from $\mathbf{b}$ to $\mathbf{a}$. Thus, instead of one edge between two states, we have two directed edges, one in each direction. Below we see a comparison of the 3-dimensional hypercube and the graph of a 3 player game with 2 actions.

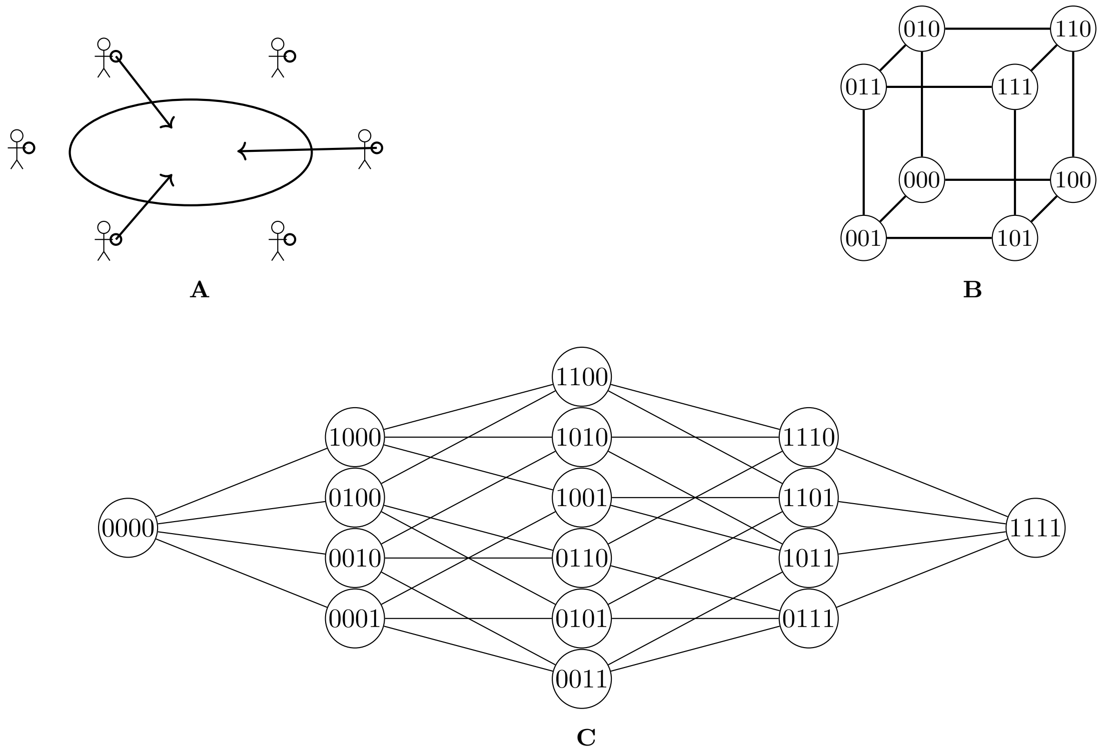


Figure 1: **A**: The public goods game with 8 players. Here 4 players contribute. **B**: The graph of a 3-player game with 2 actions under a birth-death process. This is closely related to the hypercube of dimension 3. **C**: The graph of a 4-player game with 2 actions under a birth-death process. This representation shows how one player changes action at a given time.

2.2 Population Dynamics

We adapt four established dynamics to the heterogeneous setting: the Moran process [26], Fermi imitation dynamics [40, 23, 19], aspiration dynamics [6], and introspection dynamics [4, 5]. We also introduce a novel fifth dynamic, *Introspective Imitation Dynamics*, which combines the fitness-proportionate role-model selection of the Moran process with the self-referential evaluation step of introspection dynamics.

Choice and Selection intensity

In Moran-process-type dynamics, which model biological evolution, we introduce a parameter ϵ referred to as *selection intensity* via the mapping

$$f_i(\mathbf{a}) = 1 + \epsilon\pi_i(\mathbf{a}).$$

Appropriate bounds on ϵ ensure that the fitness values f_i remain non-negative, which is necessary for the reproduction probabilities in the Moran process to be well-defined. Smaller values of ϵ correspond to weaker selection, placing less emphasis on payoff differences in the evolutionary update. In the limiting case $\epsilon = 0$, the process reduces to neutral drift, in which reproduction is independent of payoff. In contrast, population dynamics based on the Fermi update rule [40] employ a parameter β, which we refer to as the *choice intensity.* This parameter determines how sensitively individuals respond to payoff differences when making decisions. Depending on the specific dynamic under consideration, these payoff differences may arise from comparing an individual's current payoff with that of another individual in the population, or from comparing their current payoff with the payoff they would obtain by switching strategy.

The Moran Process

The Moran process [26] is defined by the following algorithm:

1. A player is chosen to be copied with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to change their strategy

The process goes from state to state according to the entry $T_{\mathbf{ab}}$ of a transition matrix T in which each row and column correspond to a state in S is defined as the probability of a transition $\mathbf{a} = (a_1, a_2, \dots, a_N) \to \mathbf{b} = (b_1, b_2, \dots, b_N)$. For the Moran process, this is as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{\sum_{a_i = b_{I(\mathbf{a},\mathbf{b})}} f(a_i)}{N \sum_{a_j} f(a_j)} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a})\text{, differing at position } I(\mathbf{a},\mathbf{b}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{1}$$

An important case which proceeds from (1) is that of a transition $\mathbf{a} \to \mathbf{b}$ where $\mathbf{b}$ contains an individual of a *type* not found in $\mathbf{a}$. For example, the transition $(0, 1) \to (0, 2)$. This would be forbidden by the intuition of a standard Moran process, as the new individual is the duplication of another individual in $\mathbf{a}$. However, we can see that the standard formula yields $T_{ab} = 0$ because there does not exist $a_i \in \mathbf{a} : a_i = b_{i^*}$.

We can now discuss one of the key ideas in the Moran process. As no new action types can be introduced to the population once they are fully removed, we have a set of absorbing states $\mathbf{a}$ such that $a_1 = a_2 = ...$, for which we have $p_{\mathbf{a} \to \mathbf{a}} = 1$. We denote the set of absorbing states S^ω. We therefore have that the Moran process forms an absorbing Markov chain [37], and using standard results can investigate the probability of the population being absorbed into each absorbing state.

Fermi Imitation Dynamics

The Fermi Imitation Dynamic is another method of modelling evolution in a population. It models the following process:

- Two players, i and j are selected at random. Unlike the original formulation [40], player j need not be a neighbour of player i, reflecting the well-mixed setting considered here.
- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})))}}$

Where β is the choice intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [23, 19, 14, 40].

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. However, Fermi Imitation Dynamics compared the fitness after the selection of players. This means that in Fermi Imitation Dynamics, there is not necessarily a player changing action in each timestep, whereas in a Moran process, there is always a player changing action type (although it may be a change to the same action type that they are currently performing) We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} (\frac{1}{N(N-1)} \sum_{a_j = b_{I(\mathbf{a},\mathbf{b})}} \phi(\pi_{\text{I}(\mathbf{a},\,\mathbf{b})}(a) - \pi_j(\mathbf{a}))) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{2}$$

Here we see how the Fermi function can be used to define the transition probability between states in a Markov chain. Importantly, we see that the first case of equation 2 is always in the interval $(0, 1)$, as the Fermi function always takes a value on $(0, 1)$, and the summation term is at most $N - 1$ values of said function, and thus takes a value $< N - 1$.

Aspiration Dynamics

Aspiration dynamics models an update rule based on how far away a player's performance in a game is from an aspirational constant fitness value:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A_i)$

Here, A_i is player i's *aspiration*, meaning the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is almost always defined for two actions, traditionally cooperation and defection [6].

The process transitions between states according to:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\,\phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_{I(\mathbf{a},\mathbf{b})}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a},\mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{3}$$

Introspection Dynamics

In introspection dynamics [4], individuals are selected to consider if a change of their action choice would lead to an increase in payoff. The new action is selected with a probability that depends on the potential increase and the selection intensity β:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. Player i chooses another possible strategy, $a_{i,l}$, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $p_{a_{I(\mathbf{a},\mathbf{b})} \to b_{I(\mathbf{a},\mathbf{b})}}(a_{-I(\mathbf{a},\mathbf{b})}) = \phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$, where $\Delta\pi_{I(\mathbf{a},\mathbf{b})} = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$ is the difference between the fitness of the player's current strategy and the possible fitness of the new strategy.

This process is well suited to a heterogeneous population. This is because in some cases using imitation dynamics, a player may copy a strategy due to its fitness when played by another individual in the population, despite the fact that such a strategy actually has a lower payoff for the focal player than their current strategy. In introspection dynamics, players do not consider the fitness of other individuals, and so cannot be fooled by high fitness strategies with attributes different to that of the focal player.

The process transitions between states according to:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(m_j - 1)}\,\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a},\mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{4}$$

where m_j is the number of actions available to individual j. Since we assume that all individuals have the same action set with size k, we can replace $m_j - 1$ with $k - 1$. The explicit transition matrix for the two-player case is given in equation (**??**) in the supplementary material.

Introspective Imitation Dynamics

Introspection dynamics allows players to learn based on their own fitness, resulting in a reduced probability of state changes that worsen the focal individual's fitness in a heterogeneous population. Now we consider a population dynamic that utilises this property of introspection dynamics but within the context of an imitative process. Consider a population of N players with the same action set. We define the introspective imitation process as follows:

1. A focal player is chosen at random to reconsider their strategy
2. Another player a_j in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered
3. The focal player changes their strategy with probability $\phi(\Delta(f_{I(\mathbf{a},\mathbf{b})}))$

We define the transition matrix T of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \frac{\sum_{a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{a},\mathbf{b})})) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{5}$$

We see that players can only copy the actions present in the state, however they determine whether or not to copy a player based on their own fitness rather than by considering the fitness of another player. This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themself just because it worked for another player. Instead, players look to others to see what action they *could* play, but judge based on their own fitness to decide what action they *should* play.

Classification of dynamics

Figure 2, shows a diagrammatic representation of the population dynamics shown in this section. It also proposes a classification our population dynamics into one of two categories. The first are population dynamics which accept strategies from the population based on the fitness of said strategies when played by other individuals. These dynamics are poorly suited to a heterogeneous population, as individuals do not consider how the change will affect them. However, these do offer a good model for biological processes such as the evolution of traits in nature, as in these cases the fitness of an individual causes them to pass on their traits. The second are population dynamics which consider their own change in payoff in the case that they accept a new strategy. These dynamics are more well suited to a heterogeneous population, modelling state changes where individuals make an active decision to change state in order to improve their own fitness.

2.3 Mutation

Three of the dynamics in our model correspond to absorbing Markov chains, where the only relevant outcome is the probability of absorption into a given absorbing state. The remaining two dynamics are ergodic chains, for which we can compute the long-run probability of cooperation. To enable meaningful comparison across all models, we introduce mutation into our framework.

Mutation ensures that all processes become ergodic, while also introducing a controlled amount of noise: at any moment, individuals may switch to a different strategy. [9]

Given an original process and two states $\mathbf{a}$ and $\mathbf{b}$, the probability of the original process *with mutation* transitioning from $\mathbf{a}$ to $\mathbf{b}$ is given by:

$$P(\mathbf{a} \to \mathbf{b}) = P(\text{pick individual } I(\mathbf{a},\mathbf{b})) \cdot \left(P(\mathbf{a} \to \mathbf{b}) \cdot P(\text{no mutation}) + P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} \text{ mutates} \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})})\right)$$

We assume a mutation parameter $\mu \in [0,1]_{\mathbb{R}}^{N \times k}$, where μ_{ij} denotes the probability that individual i adopts strategy j via mutation. For a population process with transition matrix T, the mutated process is described by $T^{(\mu)}$. In this process, a transition from state $\mathbf{a}$ to $\mathbf{b}$ occurs with probability $T_{\mathbf{ab}}$; mutation then acts on the resulting state with probability dictated by μ.

For a focal individual i, the probability that no mutation occurs is $1 - \sum_{j=1}^{k} \mu_{ij}$. The transition matrix $T^{(\mu)}$ for the mutated process is:

$$T^{(\mu)}_{\mathbf{ab}} = \begin{cases} T_{\mathbf{ab}} \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{6}$$

Note that if $\sum_{j=1}^{k} \mu_{ij} = 1$, mutation fully determines the behavior of individual i.

This construction ensures that the process is ergodic, rather than absorbing. Consequently, we no longer obtain absorption matrices for models with mutation; instead, we derive a unique steady state that is independent of the initial state.

Detailed transition matrices for each of the previously studied processes, now incorporating mutation, are provided in the supplementary materials.

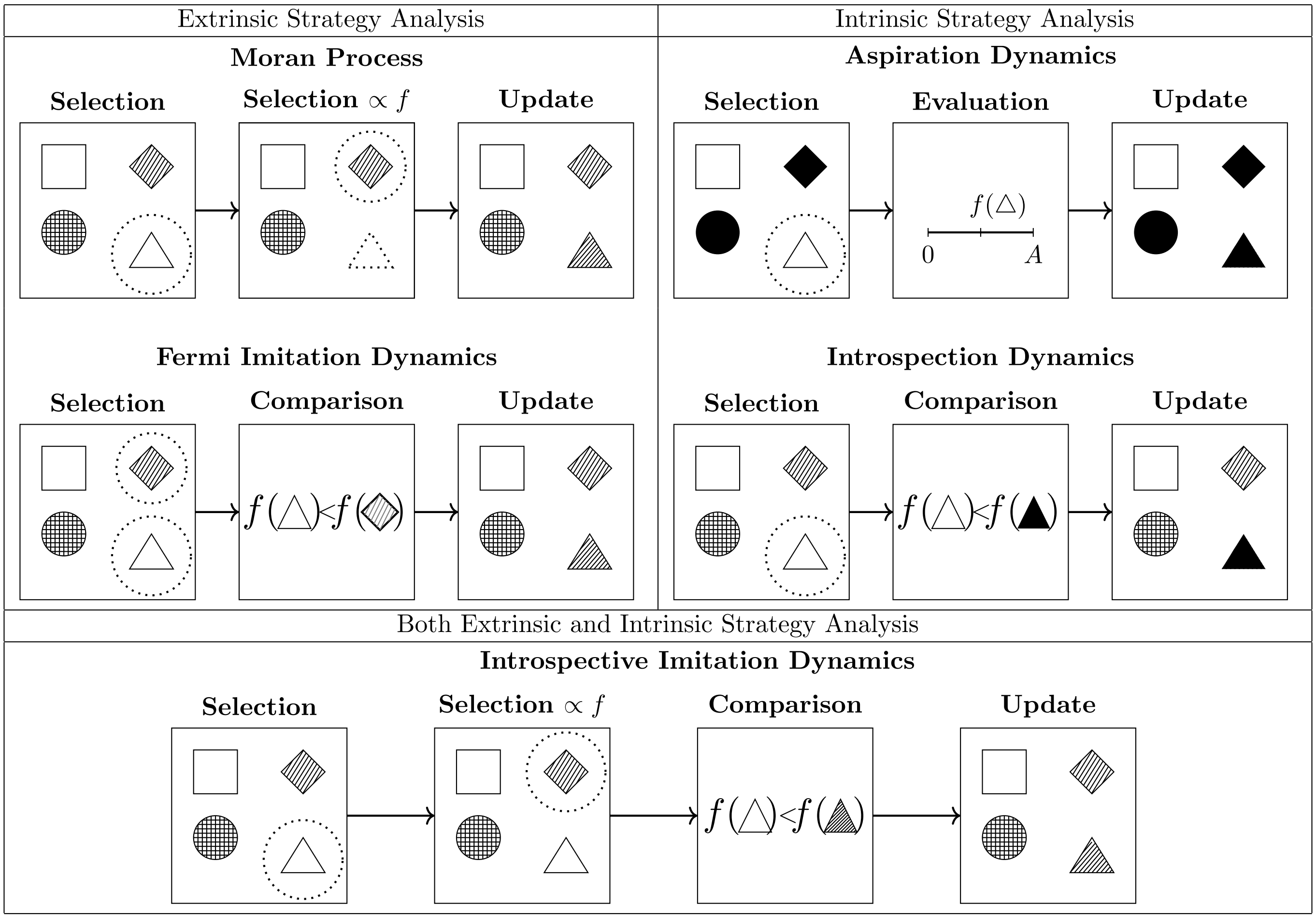


Figure 2: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability $\frac{1}{N}$ and the diamond is chosen with a varying probability depending on the process. In the case of Introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other strategy to accept.

2.4 The Public Goods Game

To illustrate all of the dynamics of Section 2 we will use the Public Goods game.

We apply the dynamics above to the public goods game [36, 13, 2, 1, 21]. In this game, individuals choose whether or not to contribute to a common resource pool. The total contribution is multiplied by a factor $r > 1$ and distributed equally among all N players. The model captures the temptation to free-ride [11]: a player who withholds their contribution still receives an equal share of the multiplied pool, so individual incentives push against the collectively optimal outcome of universal cooperation.

We can model a public goods game by providing the following payoff function:

$$\pi_i(\mathbf{a}) = \frac{r\sum_{j=1}^{N}\alpha_j}{N} - \alpha_i \tag{7}$$

Where α_j is the contribution by individual j. This is a similar heterogeneous public good game as the one described in [13] although in that model the return r is also a heterogeneous property of each player.

3 Results

3.1 Numeric Results

For the purposes of this section, a large scale parameter sweep is carried out. To the authors knowledge this is the largest such parameter sweep. It is done to the highest standards of reproducibility [44, 32, 45, 38, 41] and all data has been archived and is a further contribution of the work. For the purpose of this research, the authors developed a software library `ludics` (written in python) which is installable. The archive of the software is at [?] and the documentation can be found at `https://hefos.github.io/ludics/`. All the development of the software can be found at `https://github.com/hefos/ludics`.

Parameter Space

When we generate the data required for empirical analysis of the public goods game, we consider two popoulation structures, each defined by the value M, the maximum total value of contribution to the public good, achieved when all players contribute resources. In such a case, the payoff of all players is $\pi_i = \frac{rM}{N} - \alpha_i$.

We consider a “linear” population, in which a player in position i contributes $\frac{2Mi}{N(N+1)}$. 3 shows the combinations of parameters considered across each of the population dynamics.

Parameter	Minimum	Maximum	Number of Values per N
N	3	8	N/A
r	$\frac{1}{2}$	$\frac{N}{2}$	30
μ	0.001	0.1	4
ϵ	0	$\frac{19}{20\cdot\max_i\alpha_i}$	30
β	0	2	30
A	1	$\frac{1.01\cdot M}{N}$	50
n	1	$N-1$	$N-1$
α_h	$\frac{M}{N}$	$\frac{19\cdot M}{20(N-n)}$	30

Figure 3: The set of measured parameters and boundry values. Some parameters are only applicable to certain populations and processes, for example α_h is not a parameter of a linear population structure

Measures of Cooperation

We define p_C to be the *probability of cooperation*. This represents the mean proportion of cooperators at a given step once an evolutionary game reaches it's steady state, and can be calculated by the value $\frac{\sum_{\mathbf{a}\in S}\left(\Pi_a \sum_{i=1}^{N}\mathbf{a}_i\right)}{N}$, where $\mathbf{a}_i$ takes the value 1 if player i cooperates in state $\mathbf{a}$, and 0 if not. A special case of p_C is in an extrinsic population dynamic where $\mu = 0$. In such a case, our dynamic forms an absorbing Markov chain, and as such rather than a steady state we obtain an absorption matrix which depends on the initial state of the system. In this case, p_C can be interpreted as the probability that, given an initial state, the chain will be absorbed into the state where all players cooperate. Figure **??**

shows the marginal distribution of the cooperation probability p_C across all parameter combinations and dynamics. Most configurations yield low cooperation, but the right tail is non-trivial: meaningful cooperation is achievable under a subset of conditions. We also see the distribution of p_C across each population dynamic.

Figure 5 shows that under Fermi imitation dynamics, greater heterogeneity decreases p_C in the linear case. Higher values of α_i increase the payoff gap between cooperators and defectors within each state, making it more likely that a defector will be copied and less likely that a high contributor will persist.

Under the Moran process, the degree of linear heterogeneity has no discernible effect on p_C (Figure 5): every violin across the heterogeneity deciles is centred at the same median and has the same spread. The same invariance holds for Introspective Imitation Dynamics under linear population structures (see supplementary material).

3.2 The cooperation ceiling under purely extrinsic dynamics

Both extrinsic dynamics exhibit an upper bound on cooperation that cannot be exceeded by any parameter choice. When $\alpha_i > 0$ for all i, a contributing individual always earns less than a defector in the same state: the contributor pays α_i while receiving only $r\alpha_i/N$ of it back (since the multiplied pool is split equally), so a defector strictly benefits from the contributor's effort without bearing its cost. In a purely extrinsic dynamic, where strategy revision compares players' current payoffs directly, the probability of adopting a cooperating strategy is therefore always lower than the probability of adopting a defecting one. Cooperation cannot exceed the neutral-drift baseline:

Definition 1 (Extrinsic population dynamic)**.** *Let T be the transition matrix of a population dynamic on a state space S. We say that T defines a* purely extrinsic *process if for every $\mathbf{a}, \mathbf{b} \in S$ with $\mathbf{b} \in Neb(\mathbf{a})$, $T_{\mathbf{ab}}$ satisfies:*

1. *$T_{\mathbf{ab}}$ is a function of payoffs evaluated in $\mathbf{a}$ and in no other state.*
2. *Let $\mathcal{R}$ denote the set of players j such that $j \neq I(\mathbf{a}, \mathbf{b})$ and $\mathbf{a}_j = \mathbf{b}_{I(\mathbf{a},\mathbf{b})}$. Then $T_{\mathbf{ab}}$ is non-decreasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \in \mathcal{R}$ (and strictly increasing in at least one such j if $\mathcal{R} \neq \emptyset$), and non-increasing in the payoffs $\pi_j(\mathbf{a})$ for all $j \notin \mathcal{R}$.*
3. *T_{ab} contains no term dependent on $a_{I(\mathbf{a},\mathbf{b})}$ or $b_{I(\mathbf{a},\mathbf{b})}$, except in the payoff function $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$, mutation parameters μ_{ij}, and in the definition of the set $\mathcal{R}$.*
4. *If all players share the same payoff in state $\mathbf{a}$ (that is, there exists $v \in \mathbb{R}$ such that $\pi_i(\mathbf{a}) = v$ for every player i), then $T_{\mathbf{ab}}$ is independent of the common value v.*
5. *Let T^{nd} denote T with all payoffs set to a common value, well-defined by (4). Order $\{C, D\}^N$ componentwise with $D < C$. For every $\mathbf{a} \leq \mathbf{a}'$, there exist random variables $\mathbf{X}, \mathbf{X}'$ on a common probability space, with $\mathbf{X} \sim T^{\mathrm{nd}}(\mathbf{a}, \cdot)$ and $\mathbf{X}' \sim T^{\mathrm{nd}}(\mathbf{a}', \cdot)$, such that $\mathbf{X} \leq \mathbf{X}'$ almost surely.*

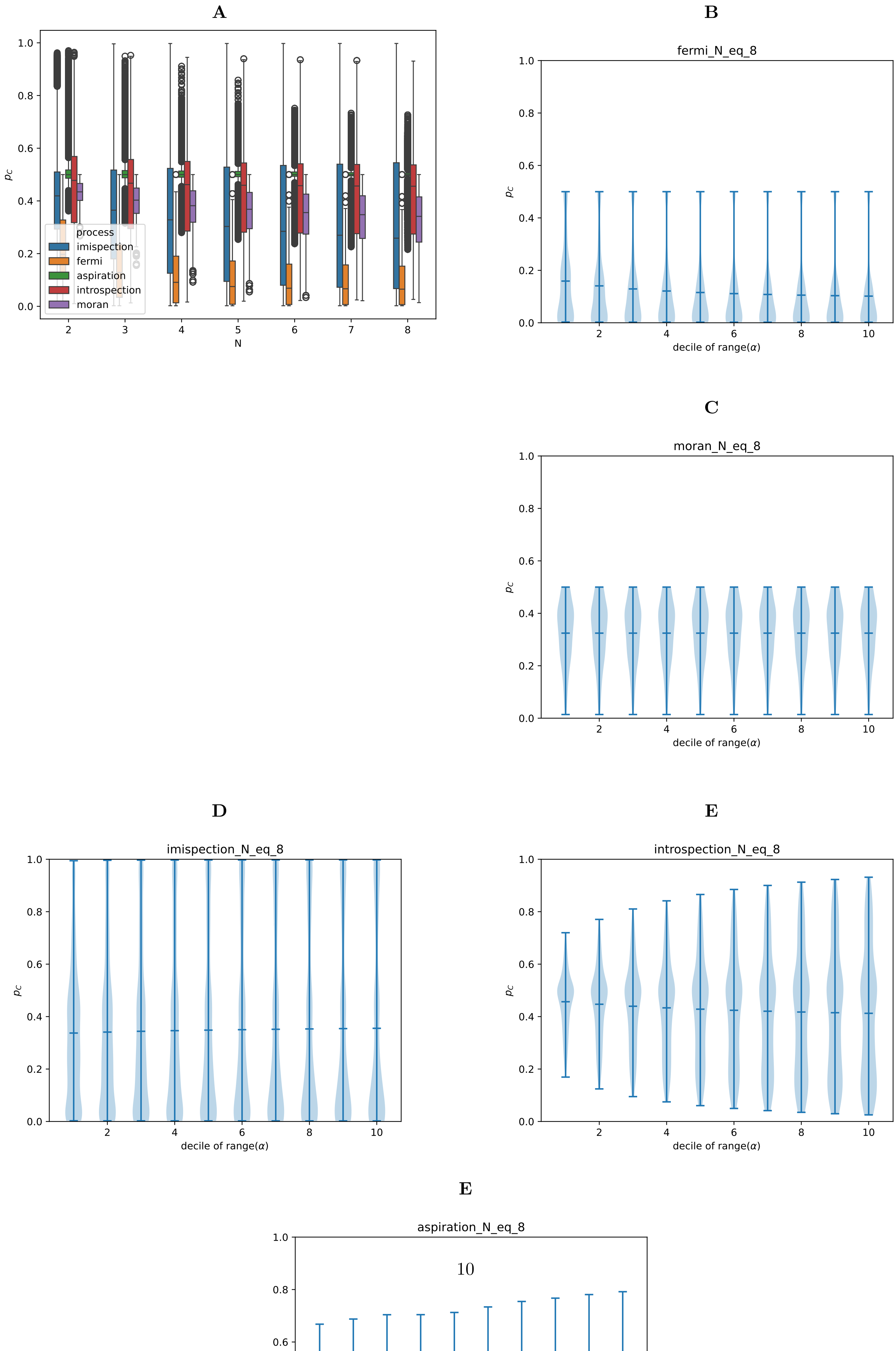
Intuitively, a purely extrinsic process revises strategies through payoff comparison. Condition (2) captures the basic monotonicity: a higher payoff for a role model makes that role model more likely to be copied, and a higher payoff for a non-role-model pulls the other way. Condition (3) rules out an intrinsic preference for either action. Conditions (4) and (5) work as a pair, ruling out hidden biases that survive when no payoff comparison is available.

Condition (4) captures payoff-scale neutrality: when all players earn the same payoff, the absolute level of that common value cannot influence $T_{\mathbf{ab}}$. Condition (5) captures configuration neutrality: under those same neutral payoffs, starting from a more cooperative profile cannot make the next state any less cooperative. Put simply: at neutral payoffs, a population that starts at least as cooperative as another remains at least as cooperative one step later. Without (5), a process could satisfy (1)–(4) and still embed a configuration-dependent pull toward defection at neutral payoffs, which is an intrinsic bias rather than truly neutral behaviour.

Note: Condition (3) in the above definition serves to exclude population dynamics which explicitly favour a certain action type, for example:

$$\bar{T}_{\mathbf{ab}} = \begin{cases} \delta_{I(\mathbf{a},\mathbf{b})} \cdot T_{\mathbf{ab}} & \text{if } \mathbf{b} \neq \mathbf{a}, \\ 1 - \sum_{\mathbf{c} \neq \mathbf{a}} T_{\mathbf{ac}} & \text{if } \mathbf{b} = \mathbf{a}. \end{cases}$$

where $T_{\mathbf{ab}}$ is the transition matrix of the some purely extrinsic dynamic, and $\delta_{I(\mathbf{a},\mathbf{b})}$ takes the value 1 if $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\frac{1}{2}$ if not. This population dynamic would be a purely extrinsic population dynamic, but would be tailored to favour a particular action type. We exclude cases such as this, as they do not conform to the framework in which fitness, is the single measure of the success of action types in the population, and thus of which strategy is passed on in each step of the process. While parameters such as choice intensity and selection intensity affect the transition probability $T_{\mathbf{ab}}$, they do not inherently favour any specific action type.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Theorem 1 (The Moran process is a purely extrinsic process)**.**

Proof. By definition the Moran process satisfies conditions (1) and (3). It remains to be shown that $T_{\mathbf{ab}}$ is non-decreasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$ (strictly increasing for at least one such j) and non-increasing in $\pi_j(\mathbf{a})$ for $j \notin \mathcal{R}$. Recall $f_i(\mathbf{a}) = 1+\epsilon\pi_i(\mathbf{a})$ for some selection intensity $\epsilon > 0$. Then we have:

$$T_{\mathbf{ab}} = \frac{\sum_{j\in\mathcal{R}} f_j(\mathbf{a})}{N\sum_{l=1}^{N} f_l(\mathbf{a})}(1-\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

$$\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = \frac{\epsilon\sum_{x\notin\mathcal{R}} f_x(\mathbf{a})}{N(\sum_{l=1}^{N} f_l(\mathbf{a}))^2}(1-\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}) \quad \text{for } j \in \mathcal{R}$$

For $j \in \mathcal{R}$, as ϵ is chosen such that $f_i(\mathbf{a}) > 0$ for all $i \in \mathbf{a}$, and $\mu_{i,j}$ such that $\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i} < 1$, this derivative is strictly positive. For $j \notin \mathcal{R}$, the analogous computation gives $\frac{dT_{\mathbf{ab}}}{d\pi_j(\mathbf{a})} = -\frac{\epsilon\sum_{x\in\mathcal{R}} f_x(\mathbf{a})}{N(\sum_{l=1}^{N} f_l(\mathbf{a}))^2}(1-\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}) \leq 0$, so condition (2) holds. Finally, for condition (4), setting $\pi_i(\mathbf{a}) = v$ for every player i gives $f_i(\mathbf{a}) = 1+\epsilon v$ for every i, whence $T_{\mathbf{ab}} = \frac{|\mathcal{R}|(1+\epsilon v)}{N\cdot N(1+\epsilon v)}(1-\sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N} = \frac{|\mathcal{R}|}{N^2}(1-\sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, which is independent of v.

For condition (5), specialise to two actions $\{C, D\}$ with $D < C$. Under T^{nd}, the dynamic samples a focal player i uniformly, a role model k uniformly from the population (since all fitnesses are equal), and a mutation outcome m from the action-invariant mutation distribution; player i then adopts a_k, or m if a mutation occurs. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by (i, k, m), applying the same draw to both. Coordinate i in the successor of $\mathbf{a}$ is a_k (or m); coordinate i in the successor of $\mathbf{a}'$ is a'_k (or the same m). Since $\mathbf{a} \leq \mathbf{a}'$ gives $a_k \leq a'_k$, and every other coordinate j is left unchanged with $a_j \leq a'_j$, the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, so condition (5) holds. □

Theorem 2 (Fermi imitation dynamics is a purely extrinsic process)**.**

Proof. By definition, Fermi imitation dynamics satisfies conditions (1) and (3). It remains to show that it satisfies conditions (2), (4), and (5).

$$T_{\mathbf{ab}} = \frac{1}{N(N-1)}\sum_{j\in\mathcal{R}} \phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_j(\mathbf{a}))(1-\sum_{i=1}^{k}\mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$$

As ϕ is strictly decreasing, every term in the sum is strictly increasing in $\pi_j(\mathbf{a})$ for $j \in \mathcal{R}$. For $j \notin \mathcal{R}$ with $j \neq I(\mathbf{a},\mathbf{b})$, $\pi_j(\mathbf{a})$ does not appear in the formula at all. For $j = I(\mathbf{a},\mathbf{b})$ (when $a_{I(\mathbf{a},\mathbf{b})} \neq b_{I(\mathbf{a},\mathbf{b})}$, so $I \notin \mathcal{R}$), each term is strictly decreasing in $\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a})$ because ϕ is. Thus, Fermi imitation dynamics satisfies condition (2). Finally, condition (4) holds: if $\pi_i(\mathbf{a}) = v$ for every player i, then every payoff difference appearing in the Fermi formula is zero, so $\phi(\pi_I(\mathbf{a}) - \pi_j(\mathbf{a})) = \phi(0) = \frac{1}{2}$ in each summand, and $T_{\mathbf{ab}} = \frac{|\mathcal{R}|}{2N(N-1)}(1-\sum_i \mu_{I(\mathbf{a},\mathbf{b}),i}) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),b_{I(\mathbf{a},\mathbf{b})}}}{N}$, independent of v.

For condition (5), specialise to two actions $\{C, D\}$ with $D < C$. Under T^{nd}, the dynamic samples a focal player i uniformly, a role model k uniformly from the remaining $N-1$ players, an independent Bernoulli outcome with success probability $\phi(0) = \frac{1}{2}$, and a mutation outcome m from the action-invariant mutation distribution; player i adopts a_k on a success (or m if a mutation occurs) and otherwise keeps a_i. Couple the chains started at $\mathbf{a} \leq \mathbf{a}'$ on the common probability space generated by $(i, k, \mathrm{Bernoulli}, m)$, applying the same draw to both. On a mutation step, both successors set coordinate i to the same m. On a successful imitation step, they set coordinate i to a_k and a'_k respectively, with $a_k \leq a'_k$. On a no-update step, coordinate i retains $a_i \leq a'_i$. Every other coordinate is unchanged, so the two successors $\mathbf{X}$ and $\mathbf{X}'$ satisfy $\mathbf{X} \leq \mathbf{X}'$ pointwise on this probability space, and condition (5) holds. □

Before stating the theorem we record a comparison result for Markov chains on partially ordered spaces. The result is classical and we claim no novelty; we include a short proof for the reader's convenience, since its use in the proof of the theorem below may not be familiar to the reader.

Lemma 1. *Let P and Q be irreducible, aperiodic transition matrices on a finite partially ordered set $(S, \leq)$, with stationary distributions π^P and π^Q. Suppose that*

(i) $P(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$; and

(ii) Q is stochastically monotone: $\mathbf{a} \leq \mathbf{a}'$ implies $Q(\mathbf{a}, \cdot) \preceq_{st} Q(\mathbf{a}', \cdot)$.

Then $\pi^P \preceq_{st} \pi^Q$.

Proof. Kamae, Krengel, and O'Brien [15] prove that whenever two probability measures μ_1, μ_2 on a partially ordered Polish space satisfy $\mu_1 \preceq_{st} \mu_2$, there exist random variables $X_1 \sim \mu_1$ and $X_2 \sim \mu_2$ defined on a common probability space with $X_1 \leq X_2$ almost surely (a *monotone coupling*).

Using this, we construct a coupling of the two chains by induction on t. Start with any $X_0^P, X_0^Q \in S$ satisfying $X_0^P \leq X_0^Q$. Given $X_t^P \leq X_t^Q$, hypothesis (ii) gives $Q(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$, which combined with (i) at $\mathbf{a} = X_t^P$ yields $P(X_t^P, \cdot) \preceq_{st} Q(X_t^Q, \cdot)$. Applying Kamae–Krengel–O'Brien to this pair of measures produces (X_{t+1}^P, X_{t+1}^Q) with the required marginals and $X_{t+1}^P \leq X_{t+1}^Q$ almost surely. By induction, $X_t^P \leq X_t^Q$ for every t. Since both chains are ergodic, their marginal laws converge to π^P and π^Q, and passing to the limit gives $\pi^P \preceq_{st} \pi^Q$. □

Theorem 3 (Extrinsic Ceiling)**.** *Let T define a purely extrinsic process on an N-player game with 2 actions, C and D, such that in any given state $\mathbf{a}$, $\min_{j:a_j=D} f_j(\mathbf{a}) > \max_{i:a_i=C} f_i(\mathbf{a})$ (that is, in any given state, any defector has greater fitness than any cooperator). Suppose further that the system exhibits action-invariant mutation $\mu_{iC} = \mu_{iD}$ for all players $i \in \mathbf{a}$.*

Then, $p_C \leq \frac{1}{2}$

Proof. Equip $S = \{C, D\}^N$ with the componentwise partial order, declaring $D < C$. For a transition matrix P, write $P(\mathbf{a}, \cdot) \preceq_{st} P(\mathbf{a}', \cdot)$ to mean that the one-step distribution from $\mathbf{a}$ under P is stochastically dominated by the one-step distribution from $\mathbf{a}'$ under P in this order. Let $T^{\text{nd}}_{\mathbf{ab}}$ denote the value of $T_{\mathbf{ab}}$ evaluated at $\pi_i(\mathbf{a}) = 0$ for every i; by condition (4), $T^{\text{nd}}_{\mathbf{ab}}$ is equally the value of $T_{\mathbf{ab}}$ under any other choice of common payoff.

Step 1: $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$ for every $\mathbf{a} \in S$.

Set $v(\mathbf{a}) := \min_{a_j=D} \pi_j(\mathbf{a})$. In every mixed state $\mathbf{a}$, the payoff-dominance hypothesis implies $\pi_i(\mathbf{a}) < v(\mathbf{a})$ for every cooperator i and $\pi_j(\mathbf{a}) \geq v(\mathbf{a})$ for every defector j. Consider a C-gaining transition $\mathbf{a} \to \mathbf{b}$, so $b_{I(\mathbf{a},\mathbf{b})} = C$ and $\mathcal{R}$ is the set of current cooperators. Replacing every $\pi_i(\mathbf{a})$ with $v(\mathbf{a})$ raises all role-model payoffs and lowers all non-role-model payoffs, so by condition (2),

$$T_{\mathbf{ab}} \;\leq\; T_{\mathbf{ab}}\big|_{\pi_i(\mathbf{a})=v(\mathbf{a}) \text{ for all } i}$$

with strict inequality whenever $\mathcal{R} \neq \emptyset$. By condition (4), the right-hand side equals $T^{\text{nd}}_{\mathbf{ab}}$. The analogous argument with reversed inequality applies to D-gaining transitions. Hence $T(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}, \cdot)$.

Step 2: T^{nd} is invariant under the $C \leftrightarrow D$ label swap.

Let $\sigma : S \to S$ be the map that swaps every C and D. By conditions (3) and (4), with common payoff 0, the only features of $(\mathbf{a}, \mathbf{b})$ that enter $T^{\text{nd}}_{\mathbf{ab}}$ are the mutation matrix and the set $\mathcal{R}$. Under action-invariant mutation the mutation matrix is σ-invariant; and $|\mathcal{R}|$ is σ-invariant because the set of players currently playing $b_{I(\mathbf{a},\mathbf{b})}$ in $\mathbf{a}$ has the same size as the set of players currently playing $\sigma(b_{I(\mathbf{a},\mathbf{b})})$ in $\sigma(\mathbf{a})$. Hence $T^{\text{nd}}_{\mathbf{ab}} = T^{\text{nd}}_{\sigma(\mathbf{a})\sigma(\mathbf{b})}$ for every transition, so the stationary distribution of T^{nd} is σ-invariant, giving $p_C^{(T^{\text{nd}})} = p_D^{(T^{\text{nd}})} = \frac{1}{2}$ [28].

Step 3: T^{nd} is stochastically monotone.

Condition (5) provides, for every $\mathbf{a} \leq \mathbf{a}'$, a coupling $(\mathbf{X}, \mathbf{X}')$ on a common probability space with $\mathbf{X} \sim T^{\text{nd}}(\mathbf{a}, \cdot)$, $\mathbf{X}' \sim T^{\text{nd}}(\mathbf{a}', \cdot)$, and $\mathbf{X} \leq \mathbf{X}'$ almost surely. For any upper set $U \subseteq S$, $\{\mathbf{X} \in U\} \subseteq \{\mathbf{X}' \in U\}$, so $T^{\text{nd}}(\mathbf{a}, U) \leq T^{\text{nd}}(\mathbf{a}', U)$, giving $T^{\text{nd}}(\mathbf{a}, \cdot) \preceq_{st} T^{\text{nd}}(\mathbf{a}', \cdot)$.

Applying Lemma 1 to $P = T$ and $Q = T^{\text{nd}}$ (which satisfy (i) by Step 1 and (ii) by Step 3) yields $\pi^T \preceq_{st} \pi^{T^{\text{nd}}}$. Since n_C is monotone in the componentwise order,

$$p_C^{(T)} \;=\; \tfrac{1}{N}\,\mathbb{E}_T[n_C] \;\leq\; \tfrac{1}{N}\,\mathbb{E}_{T^{\text{nd}}}[n_C] \;=\; p_C^{(T^{\text{nd}})} \;=\; \tfrac{1}{2}.$$

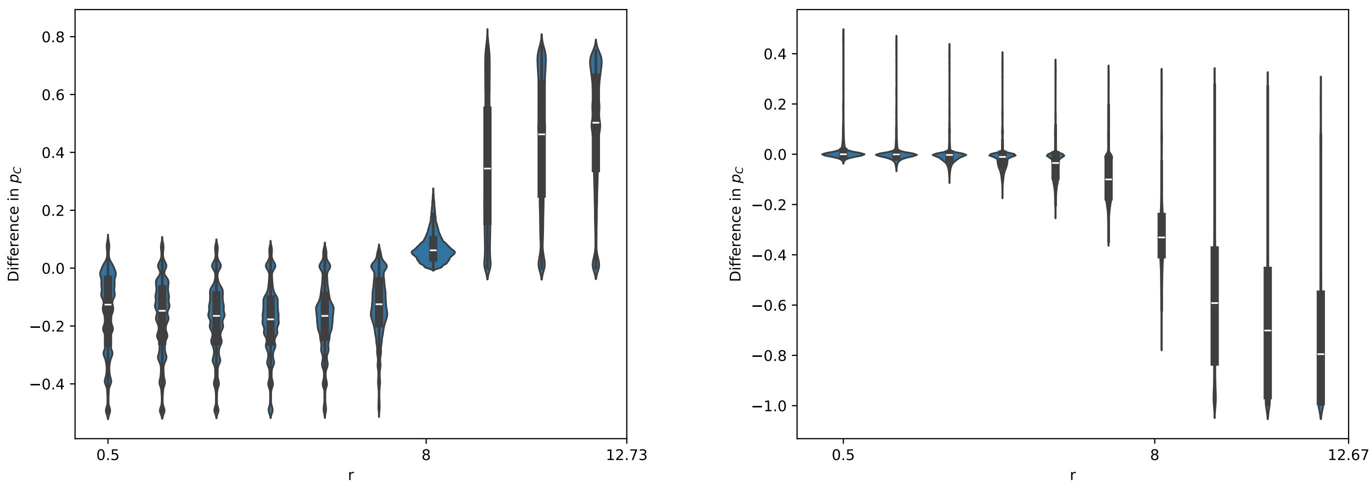
□

3.3 Intrinsic dynamics exceed the extrinsic ceiling

As seen in Figure 9, both Introspection dynamics and Introspective Imitation Dynamics break through the extrinsic ceiling. The introspective step evaluates whether the focal player's own payoff would improve under the candidate strategy, independently of whether a defecting neighbour currently outperforms cooperators. When cooperation is individually beneficial for the focal player, this step assigns a probability greater than $\frac{1}{2}$ to accepting it, enabling $p_C > \frac{1}{2}$.

The advantage of intrinsic dynamics over extrinsic dynamics holds across all values of N and both population structures examined, as Figure **??** and the extended parameter sweeps in the supplementary material confirm.

3.4 The $r = N$ threshold



(a) **A** (b) **B**

Figure 10: **A**: Comparison of p_C (IID - Moran, $N = 8$), **B**: Comparison of p_C (Fermi - IID, $N = 5$) **??**

To identify when intrinsic dynamics begin to outperform extrinsic ones, we compare p_C values from the Moran process and Introspective Imitation Dynamics directly; the latter is the Moran process augmented with a single introspective step, making the comparison clean.

The parameter $r = N$ marks the precise threshold at which intrinsic dynamics begin to favour cooperation over defection.

Proposition 1 ($r = N$ threshold)**.** *For any heterogeneous population with $\alpha_i > 0$ for all i, the introspective evaluation step of Introspective Imitation Dynamics favours cooperation if and only if $r > N$.*

Proof. Individual i evaluates switching from defection to cooperation by comparing their payoff $\pi_i(\mathbf{a})$ in the current state to the payoff $\pi_i(\mathbf{b})$ they would receive in the state where they cooperate and all others remain unchanged. Using the convention $\Delta\pi_i = \pi_i(\mathbf{a}) - \pi_i(\mathbf{b})$,

$$\Delta\pi_i = \alpha_i - \frac{r\alpha_i}{N} = \alpha_i\left(1 - \frac{r}{N}\right).$$

Since $\alpha_i > 0$, this is negative if and only if $r > N$. Since ϕ is a decreasing function, $\phi(\Delta\pi_i) > \frac{1}{2}$ precisely when $\Delta\pi_i < 0$, i.e. when $r > N$, and $\phi(\Delta\pi_i) < \frac{1}{2}$ when $r < N$. □

When $r < N$, the introspective step actively suppresses cooperation relative to the extrinsic baseline. This is visible in Figure **??** as a region where Introspective Imitation Dynamics underperforms the Moran process. The pattern holds across all values of N (see supplementary material).

A further empirical signature of the $r = N$ threshold is a sharp collapse of variance. In Figure **??**, which shows $p_C(\text{IID}) - p_C(\text{Moran})$ across the full r range, the interquartile range narrows to nearly zero precisely at $r = N$, before widening again for larger r. This variance minimum coincides exactly with the theoretical threshold, providing an unusually clean empirical marker for the analytical result.

3.5 Choice intensity and its interaction with dynamics type

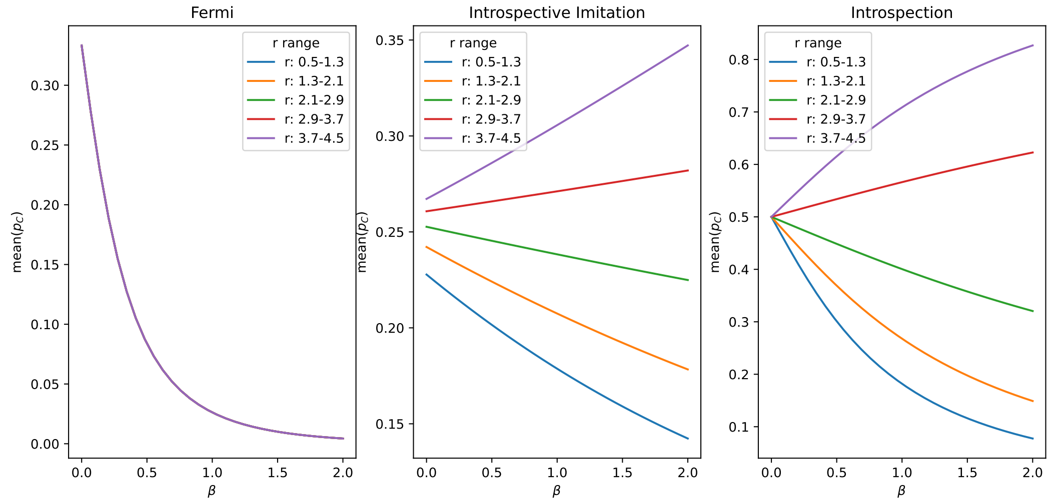


Figure 11: p_C as a function of choice intensity β and multiplier r

Higher choice intensity β consistently reduces p_C under purely extrinsic dynamics, regardless of r. A high β causes players to respond sharply to payoff differences, and since defectors always have higher payoffs than cooperators under extrinsic comparison, stronger selection accelerates the collapse of cooperation.

Under intrinsic dynamics the relationship reverses when $r > N$. The introspective step compares a player's current payoff against their counterfactual payoff under cooperation; when $r > N$ this comparison favours cooperation, and higher β amplifies that advantage, raising p_C.

Introspection dynamics has a further notable property at $\beta = 0$: all lines in Figure 11 converge to $p_C = 0.5$, independently of r or population structure. At zero choice intensity players update their strategy uniformly at random, so the stationary distribution places equal weight on all states; since exactly half of all states have the focal player cooperating, $p_C = 0.5$ follows immediately. This neutral-drift baseline therefore serves as an anchor: introspection raises p_C above 0.5 when $r > N$ and lowers it below 0.5 when $r < N$, always converging to 0.5 at $\beta = 0$.

3.6 Dominance relations between dynamics

Having examined each dynamic in isolation, we now compare them directly to establish a dominance ordering in terms of p_C.

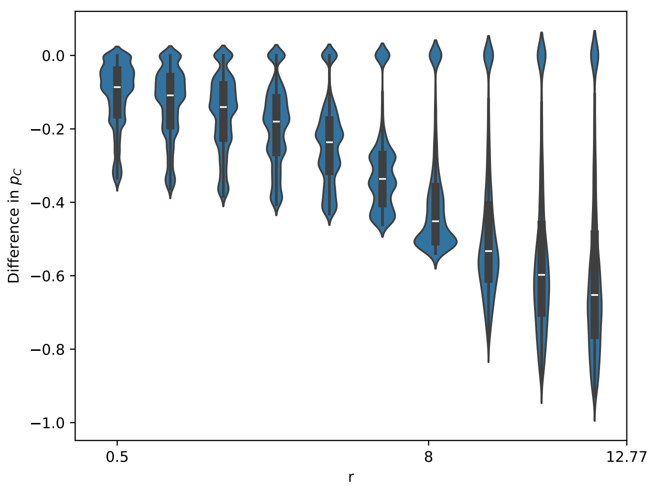
Introspection always outperforms Fermi. Figure **??** shows $p_C(\text{Introspection}) - p_C(\text{Fermi})$ across the full parameter sweep. The lower tail of every violin sits at or above zero: introspection dynamics *uniformly dominates* Fermi imitation dynamics for all parameter combinations examined.

This is due to the difference in $\Delta(f)$ in Introspection dynamics and Fermi imitation dynamics. We have the following values of Δ for a cooperator accepting a defecting strategy:

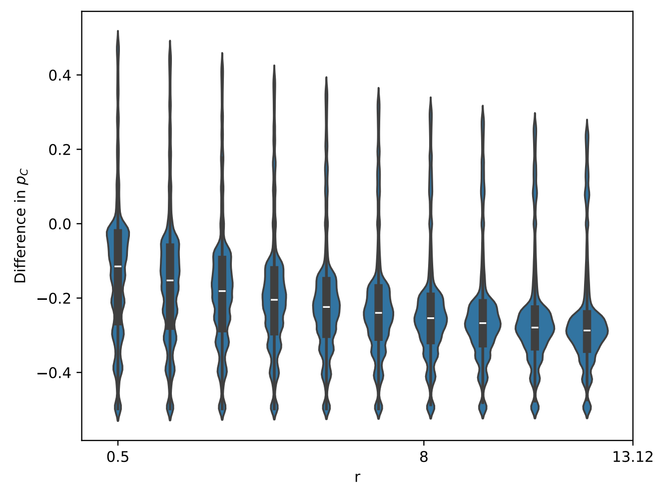
$$\Delta^{\text{Intro}}(f) = (\frac{r}{N} - 1)\alpha_i > -\alpha_i = \Delta^{\text{Fermi}}(f)$$

The inequality is reversed for a defector accepting a cooperative strategy, as each Δ is negated. As ϕ is a decreasing function, we therefore have that Fermi dynamics accept cooperation with a lower probability and accept defection with a higher probability. Since both Fermi imitation dynamics and Introspection dynamics are stochastically monotone, we have by the Kamae-Krengel-O'brien theorem [15] that $p_C^{\text{Fermi}} \leq p_C^{\text{Intro}}$ for any parameter set.

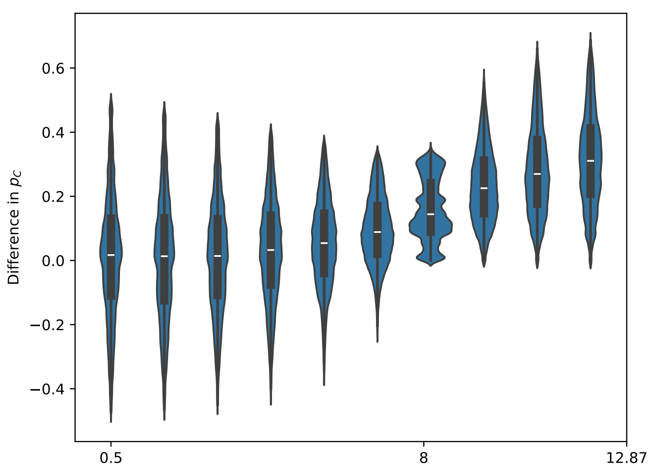
Introspection's can outperform Moran for all r Unlike introspective imitation dynamics, which has a hard line at $r = N$ to begin outperforming the Moran process, Introspection dynamics can outperform the Moran process at all values of r. At all values of r, in fact, the mean value of p_C in introspection is higher than that of the Moran process, increasing with r. At the point $r = N$, we begin to see that all parameter combinations in introspection dynamics outperform their equivalent in the Moran process.



(a) **A**



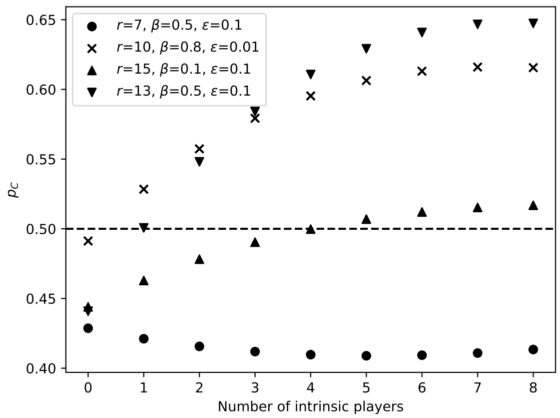
(b) **B**



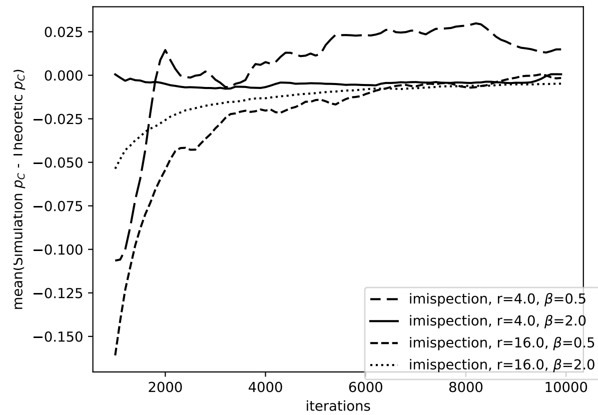
(c) **C**

Figure 12: **A**: $p_C(\text{Fermi}) - p_C(\text{Introspection})$ as a function of r for $N = 8$. **B**:$p_C(\text{Fermi}) - p_C(\text{Moran})$ as a function of r for $N = 8$ **C**: $p_C(\text{Introspection}) - p_C(\text{Moran})$ as a function of r for $N = 8$ (linear population).
??

4 Conclusion



(a) **A**



(b) **B**

Figure 13: **A**: p_C in various parameter combinations at $N = 8$, varying the number of players using an intrinsic population dynamic. **B**: for various parameter combinations, the mean value over numerous simulations of an 8 player population playing introspective imitation dynamics converging to the theoretic p_C.

We have introduced and analysed a framework for heterogeneous public goods games under five distinct population dynamics, two purely extrinsic (Moran process and Fermi imitation) and three intrinsic or mixed (aspiration, introspection, and Introspective Imitation Dynamics). Exact results are obtained by computing transition matrices and stationary distributions analytically for the two-player case, with the general-N case studied numerically.

The central theoretical finding is a dichotomy between extrinsic and intrinsic dynamics. Under any purely extrinsic dynamic, cooperation is bounded above by $p_C \leq 1/2$ (Proposition 3): payoff comparisons between players always disfavour the cooperating strategy, so the neutral-drift baseline cannot be exceeded. Intrinsic dynamics, which evaluate payoffs from the focal player's own perspective, break through this ceiling whenever $r > N$ (Proposition 1). The $r = N$ threshold is sharp: it is not only the point at which the expected difference in p_C changes sign, but also the point at which the variance in that difference collapses to a minimum.

The empirical results reveal a rich structure within and across dynamics. Introspection dynamics uniformly dominates Fermi imitation over all parameter combinations, and its advantage over the Moran process grows with r, while outperforming it on average for all values of r.

Introspective Imitation Dynamics is introduced here as a novel hybrid dynamic, combining the imitative learning structure of the Moran process with the payoff self-evaluation of introspection dynamics. It is the only absorbing dynamic that exceeds the 1/2 ceiling, and its behaviour changes qualitatively at $r = N$. We see in 13 that with just a few players updating according to introspective imitation dynamics in an otherwise extrinsic population, we can encourage p_C to exceed a the extrinsic ceiling of 1/2. Further work on this may look further into the effect of hybrid population dynamics, similar to the combination of Moran and Fermi processes in [22], where extrinsic and intrinsic population dynamics are played in a mixed population. We also show that simulations converge to the theoretic value in introspective imitation dynamics across a large number of iterations in introspective imitation dynamics. There may be further study in the use of mixing times [20] to see which population dynamics converge to their theoretic p_C in the fewest iterations.

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